

# PVLoad-R1 Bench Card

## ARDUINO → 1×10 HEADER

|    |     |          |
|----|-----|----------|
| 1  | 5V  | 5 V      |
| 2  | GND | GND      |
| 3  | D6  | K1 drive |
| 4  | D7  | K2 drive |
| 5  | D8  | K3 drive |
| 6  | D9  | CS U2    |
| 7  | D10 | CS U1    |
| 8  | D11 | SDI      |
| 9  | D12 | SDO      |
| 10 | D13 | SCK      |

Header pin 1 is the square pad. On the Uno, 5V and GND are on the power header; D8–D13 sit nearest the USB socket.

## BANANA JACKS

**J1** red is PV+. **J2** red is +24 V. **J3** black is ground, shared by the supply return and the cell negative — use a stacking plug so nothing is jammed alongside anything else.

## POWER UP

1. 24 V on
2. 5 V on (plug in the Arduino)
3. PV– → J3
4. PV+ → J1
5. illuminate the cell

## POWER DOWN

1. darken the cell
2. PV+ out of J1
3. PV– out of J3
4. Arduino out (5 V off)
5. 24 V off

**Nothing gets unplugged to switch a rail.** The two supply leads stay in J2 and J3, and the ten header wires stay between the Arduino and the board.

**24 V on / off** is the output switch on the supply. If yours has no output switch, seat the black lead in J3 first and treat the red going into J2 as the on step.

**5 V on / off** is the USB cable **at the laptop end**. Pull it there and leave the Arduino and its header wiring alone. Stop the script before you pull it.

## CABLES

**Arduino.** USB-A to USB-B, straight to the laptop.

**24 V supply.** Two banana leads to J2 and J3. Set 24.0 V, current limit about 100 mA.

**EDFA100P.** Plain USB cable, no adapter.

**The two 617s.** They have an IEEE-488 socket and nothing else, so you need a USB-GPIB adapter — National Instruments, to match the VISA on the laptop. One adapter carries both: run a cable from the adapter to the first meter, then a second cable from the first meter to the second, because the connectors stack.

Set a **different address on each meter** from its front panel before connecting them. Both ship on 27 and will collide. Use 22 for the volts meter and 23 for the amps meter.

## PUTTING THE TWO 617S IN THE CIRCUIT

Each 617's INPUT is a triaxial socket, so you cannot clip a banana lead straight on. Screw a **triax-to-dual-banana adapter** onto it. That gives you two banana posts, **HI** and **LO**.

### Amps meter — goes inside the wire

Current can only be measured by making it flow through the meter, so you cut the PV+ line and put the meter in the gap. The wire that used to run from the cell straight to J1 now runs through the meter instead:

```
cell + → HI post
LO post → J1
```

### Volts meter — sits across the cell

Nothing is cut. Two leads simply touch the cell's own two terminals:

```
HI post → cell +
LO post → cell -
```

Put these on the **cell leads themselves**, not on J1 and J3. On J1 you would be reading through the amps meter as well, which makes the voltage read low.

## THE EDFA INTERLOCK

Small connector on the back of the amplifier. It is a laser safety cutout: the amplifier only emits while that circuit is closed, the idea being that you wire it to a door switch so opening the door kills the beam. **Plug in the shorting plug it shipped with.** With nothing in it the laser will not turn on at all, no matter what the software asks for.

**LASER — Class 3B, 1550 nm, invisible.** It emits up to 30 mW whenever it is enabled, even at zero pump current and with no optical input. Terminate the output before running. Never look into the fibre bulkhead.

**WHY THE POWER ORDER MATTERS.** With 5 V up, K1 closed and 24 V missing, K1 shorts out R1 and the cell drives straight into the DigiPot's input clamp at 80% of its absolute maximum. That is what kills a chip. The cell goes on last and comes off first for that reason.